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Displayed equations

1.1 Introduction

The amsmath package provides a number of additional displayed equation structures beyond the ones provided in basic LaTeX. The augmented set includes:

```
equation     equation*     align       align*
gather       gather*       alignat     alignat*
multline     multline*     flalign     flalign*
split
```

(Although the standard `eqnarray` environment remains available, it is better to use `align` or `equation`+`split` instead. Within `eqnarray`, spacing around signs of relation is not the preferred mathematical spacing, and is inconsistent with that spacing as it appears in other environments. Long lines in this environment may result in misplaced or overprinted equation numbers. This environment also does not support the use of `\qed` or `\qedhere` as provided by theorem packages.)

Except for `split`, each environment has both starred and unstarred forms, where the unstarred forms have automatic numbering using LaTeX's `equation` counter. You can suppress the number on any particular line by putting `\notag` before the end of that line; `\notag` should not be used outside a display environment as it will mess up the numbering. You can also override a number with a tag of your own using `\tag{`⟨*label*⟩`}`, where ⟨*label*⟩ means arbitrary text such as `$*$` or `ii` used to "number" the equation. A tag can reference a different tagged display by use of `\tag{\ref{`⟨*label*⟩`}`⟨*modifier*⟩`}` where ⟨*modifier*⟩ is optional. If you are using `hyperref`, use `\ref*`; use of the starred form of `\ref` prevents a reference to a modified tag containing a nested link from linking to the original display.

There is also a `\tag*` command that causes the text you supply to be typeset literally, without adding parentheses around it. `\tag` and `\tag*` can also be used within the unnumbered versions of all the amsmath alignment structures. Some examples of the use of `\tag` may be found in the sample files `testmath.tex` and `subeqn.tex` provided with the amsmath package.

The `split` environment is a special subordinate form that is used only *inside* one of the others. It cannot be used inside `multline`, however. `split` supports only one alignment (`&`) column; if more are needed, `aligned` or `alignedat` should be used. The width of a `split` structure is the full line width.

In the structures that do alignment (`split`, `align` and variants), relation symbols have an `&` before them but not after—unlike `eqnarray`. Putting the `&` after the relation symbol will interfere with the normal spacing; it has to go before.

In all multiline environments, lines are divided by `\\`. The `\\` should *not* be used to end the last line. Using it there will result in unwanted extra vertical space following the display.

In *all* math environments (inline or display), blank lines (equivalent to `\par`) are not permitted, and will result in an error.

1.2 Single equations

The `equation` environment is for a single equation with an automatically generated number. The `equation*` environment is the same except for omitting the number.[1] The wrapper `\[ ... \]` is equivalent to `equation*`.

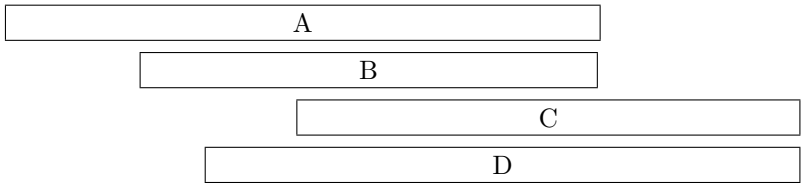
1.3 Split equations without alignment

The `multline` environment is a variation of the `equation` environment used for equations that don't fit on a single line. The first line of a `multline` will be at the left margin and the last line at the right margin, except for an indention on both sides in the amount of `\multlinegap`. Any additional lines in between will be centered independently within the display width (unless the `fleqn` option is in effect).

Like `equation`, `multline` has only a single equation number (thus, none of the individual lines should be marked with `\notag`). The equation number is placed on the last line (`reqno` option) or first line (`leqno` option); vertical centering as for `split` is not supported by `multline`.

It's possible to force one of the middle lines to the left or right with commands `\shoveleft`, `\shoveright`. These commands take the entire line as an argument, up to but not including the final `\\`; for example

(1.10)




```
\begin{multline}
\framebox[.65\columnwidth]{A}\\
\framebox[.5\columnwidth]{B}\\
\shoveright{\framebox[.55\columnwidth]{C}}\\
\framebox[.65\columnwidth]{D}
\end{multline}
```

The value of `\multlinegap` can be changed with the usual LaTeX commands `\setlength` or `\addtolength`.

1.4 Split equations with alignment

Like `multline`, the `split` environment is for *single* equations that are too long to fit on one line and hence must be split into multiple lines. Unlike `multline`, however, the `split` environment provides for alignment among the split lines, using `&` to mark alignment points. Unlike the other amsmath equation structures, the `split` environment provides no numbering, because it is intended to be used

[1] Basic LaTeX doesn't provide an `equation*` environment, but rather a functionally equivalent environment named `displaymath`.

Table 1.1. Comparison of displayed equation environments (vertical lines indicating nominal margins)

```
\begin{equation*}
a=b
\end{equation*}
```

$$a = b$$

```
\begin{equation}
a=b
\end{equation}
```

$$a = b \tag{1}$$

```
\begin{equation}\label{xx}
\begin{split}
a& =b+c-d\\
 & \quad +e-f\\
 & =g+h\\
 & =i
\end{split}
\end{equation}
```

$$\begin{aligned} a &= b + c - d \\ &\quad + e - f \\ &= g + h \\ &= i \end{aligned} \tag{2}$$

```
\begin{multline}
a+b+c+d+e+f\\
+i+j+k+l+m+n
\end{multline}
```

$$\begin{aligned} a + b + c + d + e + f \\ + i + j + k + l + m + n \end{aligned} \tag{3}$$

```
\begin{gather}
a_1=b_1+c_1\\
a_2=b_2+c_2-d_2+e_2
\end{gather}
```

$$a_1 = b_1 + c_1 \tag{4}$$
$$a_2 = b_2 + c_2 - d_2 + e_2 \tag{5}$$

```
\begin{align}
a_1& =b_1+c_1\\
a_2& =b_2+c_2-d_2+e_2
\end{align}
```

$$a_1 = b_1 + c_1 \tag{6}$$
$$a_2 = b_2 + c_2 - d_2 + e_2 \tag{7}$$

```
\begin{align}
a_{11}& =b_{11}&
  a_{12}& =b_{12}\\
a_{21}& =b_{21}&
  a_{22}& =b_{22}+c_{22}
\end{align}
```

$$a_{11} = b_{11} \qquad a_{12} = b_{12} \tag{8}$$
$$a_{21} = b_{21} \qquad a_{22} = b_{22} + c_{22} \tag{9}$$

```
\begin{flalign*}
a_{11} + b_{11}& = c_{11}&
  a_{12}& =b_{12}\\
b_{21}& = c_{21}&
  a_{22}& =b_{22}+c_{22}
\end{flalign*}
```

$$a_{11} + b_{11} = c_{11} \qquad a_{12} = b_{12}$$
$$b_{21} = c_{21} \qquad a_{22} = b_{22} + c_{22}$$

only inside some other displayed equation structure, usually an `equation`, `align`, or `gather` environment, which provides the numbering. For example:

$$H_c = \frac{1}{2n}\sum_{l=0}^{n}(-1)^l(n-l)^{p-2}\sum_{l_1+\cdots+l_p=l}\prod_{i=1}^{p}\binom{n_i}{l_i} \cdot[(n-l)-(n_i-l_i)]^{n_i-l_i}\cdot\Bigl[(n-l)^2-\sum_{j=1}^{p}(n_i-l_i)^2\Bigr]. \tag{1.11}$$

```
\begin{equation}\label{e:barwq}\begin{split}
H_c&=\frac{1}{2n} \sum^n_{l=0}(-1)^{l}(n-{l})^{p-2}
  \sum_{l _1+\dots+ l _p=l}\prod^p_{i=1} \binom{n_i}{l _i}\\
&\quad\cdot[(n-l )-(n_i-l _i)]^{n_i-l _i}\cdot
  \Bigl[(n-l )^2-\sum^p_{j=1}(n_i-l _i)^2\Bigr].
\end{split}\end{equation}
```

The `split` structure should constitute the entire body of the enclosing structure, apart from commands like `\label` that produce no visible material.

1.5 Equation groups without alignment

The `gather` environment is used for a group of consecutive equations when there is no alignment desired among them; each one is centered separately within the text width (see Table 1.1). Equations inside `gather` are separated by a `\\` command. Any equation in a `gather` may consist of a `\begin{split}` ... `\end{split}` structure—for example:

```
\begin{gather}
  first equation\\
  \begin{split}
    second & equation\\
           & on two lines
  \end{split}
  \\
  third equation
\end{gather}
```

1.6 Equation groups with mutual alignment

The `align` environment is used for two or more equations when vertical alignment is desired; usually binary relations such as equal signs are aligned (see Table 1.1).

To have several equation columns side-by-side, use extra ampersands to separate the columns:

$$x = y \qquad X = Y \qquad a = b + c \tag{1.12}$$

$$x' = y' \qquad X' = Y' \qquad a' = b \tag{1.13}$$

$$x + x' = y + y' \qquad X + X' = Y + Y' \qquad a'b = c'b \tag{1.14}$$

```
\begin{align}
x&=y        & X&=Y        & a&=b+c\\
x'&=y'      & X'&=Y'      & a'&=b\\
x+x'&=y+y' & X+X'&=Y+Y' & a'b&=c'b
\end{align}
```

Line-by-line annotations on an equation can be done by judicious application of `\text` inside an `align` environment:

$$x = y_1 - y_2 + y_3 - y_5 + y_8 - \dots \quad \text{by (??)} \tag{1.15}$$
$$= y' \circ y^* \quad \text{by (??)} \tag{1.16}$$
$$= y(0)y' \quad \text{by Axiom 1.} \tag{1.17}$$

```
\begin{align}
x& = y_1-y_2+y_3-y_5+y_8-\dots
                      && \text{by \eqref{eq:C}}\\
 & = y'\circ y^*      && \text{by \eqref{eq:D}}\\
 & = y(0) y'          && \text {by Axiom 1.}
\end{align}
```

A variant environment `alignat` allows the horizontal space between equations to be explicitly specified. This environment takes one argument, the number of "equation columns" (the number of pairs of right-left aligned columns; the argument is the number of pairs): count the maximum number of `&`s in any row, add 1 and divide by 2.

$$x = y_1 - y_2 + y_3 - y_5 + y_8 - \dots \quad \text{by (??)} \tag{1.18}$$
$$= y' \circ y^* \quad \text{by (??)} \tag{1.19}$$
$$= y(0)y' \quad \text{by Axiom 1.} \tag{1.20}$$

```
\begin{alignat}{2}
x& = y_1-y_2+y_3-y_5+y_8-\dots
                    &\quad& \text{by \eqref{eq:C}}\\
 & = y'\circ y^*     && \text{by \eqref{eq:D}}\\
 & = y(0) y'         && \text {by Axiom 1.}
\end{alignat}
```

The environment `flalign` ("full length alignment") stretches the space between the equation columns to the maximum possible width, leaving only enough space at the margin for the equation number, if present.

$$x = y \qquad X = Y \tag{1.21}$$
$$x' = y' \qquad X' = Y' \tag{1.22}$$
$$x + x' = y + y' \qquad X + X' = Y + Y' \tag{1.23}$$

```
\begin{flalign}
x&=y        & X&=Y\\
x'&=y'      & X'&=Y'\\
```

```
x+x'&=y+y' & X+X'&=Y+Y'
\end{flalign}
```

$$\begin{aligned} x &= y & X &= Y \\ x' &= y' & X' &= Y' \\ x + x' &= y + y' & X + X' &= Y + Y' \end{aligned}$$

```
\begin{flalign*}
x&=y        & X&=Y\\
x'&=y'      & X'&=Y'\\
x+x'&=y+y' & X+X'&=Y+Y'
\end{flalign*}
```